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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CS1231-1 – INTRODUCTION TO DATA SCIENCE

(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The _____ feature in Excel allows you to quickly fill cells with repetitive or sequential data by dragging the fill handle
A) AutoSum
B) DragFill
C) QuickFill
D) AutoFill
2. Referencing the cell of other worksheets is known as _____ referencing
A) Remote
B) Mixed
C) External
D) Relative
3. Predict the output of the given function =SUBSTITUTE("Python Excel Python Excel Python Excel", "Python", "MS", 3).
A) MS Excel
B) MS MS MS Excel
C) Python Excel MS Excel
D) Python Excel Python Excel MS Excel
4. Assume data in a cell A1 as 10+300+400. How do you find second occurrence of '+' symbol in cell A1?
A) =FIND("+", A1, (FIND("+", A1, 1)))
B) =FIND("+", A1, 1)
C) =FIND("+", A1, (FIND("+", A1, 1))+1)
D) =FIND(A1, (FIND("+", A1, 1)))
5. Which of the following is a key benefit of using predictive analytics?
A) Understanding past data
B) Predicting future outcomes
C) Cleaning raw data
D) Visualizing data
6. Which excel function is used to count the number of times a particular city appears in a customer list.
A) COUNT
B) SUM
C) COUNTIF
D) COUNTA
7. In the context of data cleaning, what does the excel function CLEAN() do?
A) Removes all non-printable characters from text
B) Removes all spaces from text
C) Converts text to lowercase
D) Finds and replaces text
8. If you have a list of data that you want to group and summarize, you can use excel _____ and Outline to display summary rows or columns.
A) Subtotal
B) Autosum
C) Total
D) Autofill
9. What is the purpose of establishing relationships between tables in an Excel Data Model?
A) To enable relational data analysis across multiple tables
B) To create charts
C) To format tables
D) To validate data entry

10. Which color is commonly used in KPIs to indicate that performance is meeting the target?
 A) Yellow B) Green
 C) Red D) Blue
11. What is the primary consideration when designing aesthetically pleasing Power View reports?
 A) Using consistent color schemes B) Adding as much data as possible
 C) Avoiding the use of charts D) Using different fonts for each section
12. In Excel, which function would you use to calculate the future value of an investment based on periodic, constant payments?
 A) FV B) PV
 C) NPV D) IRR
13. What is the purpose of a KPI in data analysis?
 A) To create charts B) To measure performance against specific goals
 C) To consolidate data D) To validate data
14. What does KPI stand for?
 A) Key Performance Index B) Key Position Indicator
 C) Key Performance Indicator D) Key Projection Insight
15. Which among the following is not R Data Visualization package?
 A) plotly B) ggplot
 C) matplotlib D) shiny
16. How to create horizontal bar chart in R?
 A) barplot(data, horiz = TRUE) B) plot(data, type = "h")
 C) plot(data, type = "l") D) barplot(data)
17. A _____ depicts information like the minimum and maximum data point, the median value, first and third quartile, and interquartile range.
 A) Boxplot B) Bar plot
 C) Scatter plot D) Pie plot
18. In a histogram, continuous values are grouped and displayed in _____ whose size can be varied.
 A) lists B) vectors
 C) matrices D) bins
19. A _____ is an R-object which can contain many different types of elements inside it.
 A) List B) Logical vector
 C) Vector D) Factor
20. Which function generates sequence 1, 3, 5, 7, 9 in R?
 A) seq(1, 10) B) seq(1, 10, by = odd)
 C) seq(1, 10, by = 2) D) range(1, 10, by = 2)

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain with examples any four Text Functions in Excel .	6	L*2	2	1
b)	Define the following terms with examples: i. Business Intelligence ii. Statistical Analysis iii. Predictive Analytics iv. Text Analytics	6	L1	1	1
c)	Given: Project Start Date = 01-Apr-2024 (A1), Workdays = 20, Holiday = 15-Apr-2024 (B1). (i) Find the project end date with standard weekends (Sat–Sun). (ii) Find the project end date if weekends are Fri–Sat. (iii) Modify the formula if another holiday (22-Apr-2024) is added.	4	L3	1	1

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2.	a)	List and explain any four categories in Conditional Formatting with examples.	8	L2	2	1
	b)	You are given a dataset of employees with the following fields: Employees (ID, Name, Dept, Salary). Perform the following tasks in Excel: i. Sort the dataset first by Department (A–Z) and then by Salary (Smallest to Largest) within each department. ii. Filter the dataset to show only employees in the HR Department with Salary > 50,000.	4	L3	2	1
	c)	Consider the Sales data with fields Salesperson, Region, Amount. Write and explain Excel formula to find (i) sales in "South", (ii) total sales by "Ravi".	4	L3	2	1
3.	a)	Describe the process of Subtotals in Excel with example.	6	L2	2	1
	b)	Explain four Time functions in Excel with examples.	4	L2	2	1
	c)	Given a movie dataset: Title, Director, Duration, Release Date. Write formulas to find: i. Movies ≥ 150 min. ii. Movies longer than average. iii. Directors starting with "M". iv. Directors name length = 8. v. Movies after 01-Jan-2015. vi. Movies by Christopher Nolan = 150 min.	6	L3	2	1
Unit – II						
4.	a)	Explain INDEX and MATCH with dataset examples.	6	L2	2	1
	b)	Discuss the steps and benefits of Data Consolidation in Excel.	6	L2	3	1
	c)	Explain Data Model and Relationships in Excel.	4	L2	3	1
5.	a)	What is a Pivot Table? Explain with a Product-Sales dataset.	6	L2	3	1
	b)	Discuss three types of Data Visualization in Excel.	6	L2	3	1
	c)	Write a short note on KPIs with employee dataset.	4	L3	3	1
6.	a)	Discuss What-if Analysis with example.	6	L2	3	1
	b)	Explain the process for importing CSV and SQL data into Excel and creating a Data Model.	6	L3	3	1
	c)	Explain the concept of Power Pivot in data exploration.	4	L2	3	1
Unit – III						
7.	a)	Describe four control structures in R with syntax and examples.	8	L2	4	1
	b)	Explain vectors and factors in R with examples.	8	L3	4	1
8.	a)	Explain how R creates various charts.	8	L2	5	1
	b)	Explain different types of argument matching in R with examples.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
